NEXT GEN CORP

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Project Overview

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Business Overview

Overview

Next Gen Corp is a rapidly growing technology company that specializes in creating cutting edge hardware and software solutions. In order to ensure expansion, the organization takes pride in luring top talents across the globe, and upholding excellent employee satisfaction.

Problem statement

There are growing concerns about employee turnover, salary disparities and performance variability within departments.

Analysis

- Analyse employee turnover trends and identify root causes of high turnover rates
- Analyse employee performance performance across different departments and identify areas where performance can be improved
- Analyse salary distribution and ensure fair compensation performance and departmental benchmarks

Employee Retention Analysis

1. Who are the top 5 highest serving employees?

```
--1.Who are the top 5 highest serving employees?
select employee_id, concat(first_name, '', last_name) As full_name, hire_date,
age(current_date,hire_date) as service_duration
from employee
order by hire_date
limit 5
```

	employee_id [PK] integer	full_name text	hire_date date	service_duration interval
1	8	DavidMoore	2015-06-30	10 years 1 mon 19 days
2	48	FrankSmith	2015-09-08	9 years 11 mons 11 days
3	67	JaneBrown	2015-10-12	9 years 10 mons 7 days
4	44	JohnJohnson	2015-10-27	9 years 9 mons 23 days
5	57	JohnDoe	2016-02-16	9 years 6 mons 3 days

Insights

- **Top 5 employees are David Moore, Frank Smith, Jane Brown, John Johnson, and John Doe.**
- **All top 5 employees have been employed for over 9 years, with David Moore as the longest serving employee at 10 years.**
- Majority of long staying employees were hired between 2015 and 2016

Actionable recommendations

- Conduct comprehensive stay interviews to determine key drivers for loyalty to replicate across the organization
- Analyze the 2015-2016 hiring process
- Continuous monitoring of top staying employees

2. What is the turnover rate for each department?

```
select d.department_name,
round(cast(count(t.turnover_id)as decimal)/ count(e.employee_id)* 100,2) as turnover_rate_percent
from department d
join employee e on d.department_id=e.department_id
left join turnover t on e.employee_id = t.employee_id
group by d.department_name
order by turnover_rate_percent desc
```

	department_name character varying (30)	turnover_rate_percent numeric
1	Marketing	92.86
2	Engineering	66.67
3	Sales	27.59
4	HR	27.27

Insights

- Marketing (92.86%) and Engineering (66.67%) have alarmingly high turnover rates, indicating severe underlying problems.
- Stark contrast between high turnover departments and stable ones (Sales, 27.59% and HR, 27.27%), suggesting localized issues
- Marketing's turnover is nearly 1.5x Engineering's and 3x Sales/HR.

Actionable recommendations

- Deep-dive root cause analysis i.e conduct exit/stay interviews and surveys, etc
- Targeted retention strategies in low turnover departments i.e career pathing, competitive compensation, improved work-life balance and recognition programs
- Best practice transfer; Cross departmental workshops, mentorship programs and policy review from Sale/HR

3. **Which employees are at risk of leaving based on their Performance?**

```sql
SELECT e.employee_id, CONCAT(e.first_name, ' ', e.last_name) AS full_name, d.department_name,
COUNT(p.performance_id) AS performance_review_frequency,
ROUND(AVG(p.performance_score), 2) AS avg_performance_score,
CASE
WHEN AVG(p.performance_score) <= 3.0 THEN 'High Risk'
WHEN AVG(p.performance_score) <= 3.5 THEN 'Moderate Risk'
ELSE 'Low Risk'
END AS risk_level
FROM employee e
JOIN performance p ON p.employee_id = e.employee_id
JOIN department d ON e.department_id = d.department_id
GROUP BY e.employee_id, full_name, d.department_name
HAVING AVG(p.performance_score) <= 3.5
ORDER BY avg_performance_score;
```

	employee_id integer	full_name text	department_name character varying (30)	performance_review_frequency bigint	avg_performance_score numeric	risk_level text
1	22	Grace Wilson	Sales	6	3.43	Moderate Risk

Insights

- Grace Wilson in Sales is the only employee with a moderate risk of leaving the company.
- She has an average performance score of 3.43

Actionable recommendations

- Managerial check-in
- Performance improvement plans (PIP)
- Focused mentorship plans

4. **What are the main reasons employees are leaving the company?**

```sql
SELECT t.reason_for_leaving, COUNT(*) AS number_of_exits,
    ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER (), 2) AS percentage_of_total_exits,
    STRING_AGG(DISTINCT CONCAT(e.first_name, ' ', e.last_name), ', ') AS employees_who_left
FROM turnover t
JOIN employee e ON t.employee_id = e.employee_id
GROUP BY t.reason_for_leaving
ORDER BY number_of_exits DESC;
```

	reason_for_leaving text	number_of_exits bigint	percentage_of_total_exits numeric	employees_who_left text
1	Personal	11	39.29	Alice Brown, Alice Moore, Bob Moore, Charlie Davis, Charlie Lee, Frank Davis, Frank Doe, Grace Green, Jane Brown, John Brown, John Doe
2	Found Another Job	7	25.00	Bob Johnson, Charlie Green, David Wilson, Frank Green, Frank Wilson, Grace Wilson, John Green
3	Career Growth	5	17.86	Alice Miller, Bob Lee, David Smith, Frank Miller, Grace Lee
4	Retired	5	17.86	Alice Doe, Frank Smith, Jane Miller, John Lee, John Miller

Insights

- ‘Personal’ reasons accounts for the largest proportion of exits (39.29%).
- ‘Found another Job’ (25%) highlights external market influence and company competitiveness in talent retention.
- ‘Career growth’ (17.86%) indicates potential gaps in internal advancement opportunities

Actionable recommendations

- Deep-dive into ‘Personal’ reasons for leaving, by conducting enhanced exit interviews
- Provide regular compensation review, career development opportunities and recognition programs
- Conduct regular pulse surveys, HR analytics dashboards and manager training on retention
- Cultivate a positive work environment

Performance Analysis

1. **How many employees have left the company ?**

```sql
SELECT COUNT(*) AS employees_left
FROM turnover;
```

	employees_left bigint
1	28

Insights

- 28 out of 60 employees have left the company

Actionable recommendations

- Root cause analysis i.e conduct exit/stay interviews and surveys, etc
- Targeted retention strategies in low turnover departments i.e career pathing, competitive compensation, improved work-life balance and recognition programs
- Conduct comprehensive stay interviews to determine key drivers for loyalty to replicate across the organization

2. **How many employees have a performance score of 5.0/below 3.5 ?**

```sql
SELECT COUNT(DISTINCT employee_id) AS employee_count
FROM performance
WHERE performance_score = 5 OR performance_score < 3.5
```

	employee_count bigint
1	47

Insights

- 47 number of employees scored between 5 and 3.5

Actionable recommendations

- The number of employees with a perfect score (5.0) can highlight top performers who may be candidates for promotion, special projects, or recognition programs.
- The number of employees with a low score (below 3.5) signals potential issues that need to be addressed. This could point to the need for targeted training, performance improvement plans, or further investigation into underlying challenges they might be facing.

3. **How department has the most employees with a performance of 5.0/below 3.5?**

```sql
SELECT d.department_name, COUNT(DISTINCT p.employee_id) AS employee_count
FROM performance p
JOIN department d ON p.department_id = d.department_id
WHERE p.performance_score = 5.0 OR p.performance_score < 3.5
GROUP BY d.department_name
ORDER BY employee_count DESC
LIMIT 2;
```

	department_name character varying (30)	employee_count bigint
1	Engineering	17
2	Marketing	17

Insights

- Engineering and Marketing has the number of employees with performance of 5 and below 3.5

Actionable recommendations

Conduct regular performance data reviews, to generate early warning systems and provide management training on early intervention

4. **What is the average performance score by department?**

```sql
SELECT d.department_name,
    ROUND(AVG(p.performance_score), 2) AS avg_performance_score
FROM employee e
JOIN performance p ON e.employee_id = p.employee_id
JOIN department d ON e.department_id = d.department_id
GROUP BY d.department_name
ORDER BY avg_performance_score DESC;
```

	department_name character varying (30)	avg_performance_score numeric
1	Marketing	4.17
2	HR	4.11
3	Sales	4.07
4	Engineering	4.01

Insights

- All departments show high average performance scores indicating a generally high-performing workforce
- Marketing has the highest average score (4.17), setting a benchmark for excellence
- Engineering has the lowest score (4.01), suggesting areas for targeted support
- There is very small difference in performance range (4.01 to 4.17) across all 4 departments

Actionable recommendations

- Sustain and leverage high performance from Marketing and HR through best practice sharing, leadership development and internal/ cross departmental benchmarking
- Conduct targeted performance enhancement through needs assessments, tailored skill development, performance coaching and resource optimization

Salary Analysis

1. **What is the total salary expense for the company?**

```
SELECT COUNT(employee_id) AS no_of_employees,
    TO_CHAR(SUM(salary_amount):: numeric, 'FM$999,999,999.00') AS total_salary_expense
FROM salary;
```

	no_of_employees bigint	total_salary_expense text
1	60	$4,850,000.00

Insights

- Total annual salary expense of $4,850,000 for 60 employees represents a substantial investment
- Average salary per employee of $80,833.33 suggests a competitive compensation level

Actionable recommendations

- Perform a detailed breakdown of sales compensation, career pathing, and their alignment with business goals
- Perform a comprehensive compensation review on marketing specialists
- Establish compensation bands and framework (eg. job grading)

2. What is the average salary by job title?

```
SELECT e.job_title,
    TO_CHAR(AVG(s.salary_amount):: numeric, 'FM$999,999,999.00') AS avg_salary
FROM employee e
LEFT JOIN salary s ON s.employee_id = e.employee_id
GROUP BY job_title
ORDER BY avg_salary DESC;
```

	job_title character varying (30)	avg_salary text
1	Sales Representative	$84,285.71
2	HR Specialist	$81,818.18
3	Engineer	$80,000.00
4	Sales Manager	$80,000.00
5	Marketing Specialist	$77,857.14

Insights

- Sales related roles (Sales reps and managers) have the highest average salaries
- Sales reps have higher average salary than sales managers
- HR specialists have the second highest average salary (81,818)
- Marketing Specialists have the lowest average salary(77,857)
- Engineers and sales managers have identical average salary (80,000)

Actionable recommendations

- Perform a detailed breakdown of sales compensation, career pathing, and their alignment with business goals
- Perform a comprehensive compensation review on marketing specialists
- Establish compensation bands and framework (e.g Job grading)

3. **How many employees earn above 80,000?**

```sql
SELECT COUNT(DISTINCT s.employee_id) AS high_earners
FROM salary s
WHERE s.salary_amount > 80000;
```

	high_earners bigint
1	26

Insights

- 26 employees (43.3%) earn above $80,000 annually
- This indicates a substantial investment in higher compensated roles

Actionable recommendations

- Create Higher earning growth opportunities for all employees

4. **How does performance correlate with salary across departments?**

```sql
SELECT d.department_name,
  ROUND(AVG(p.performance_score), 2) AS avg_performance_score,
  ROUND(AVG(s.salary_amount), 2) AS avg_salary
FROM employee e
LEFT JOIN performance p ON e.employee_id = p.employee_id
LEFT JOIN salary s ON e.employee_id = s.employee_id
LEFT JOIN department d ON e.department_id = d.department_id
GROUP BY d.department_name
ORDER BY avg_performance_score DESC;
```

	department_name character varying (30)	avg_performance_score numeric	avg_salary numeric
1	Marketing	4.17	77857.14
2	HR	4.11	81818.18
3	Sales	4.07	82068.97
4	Engineering	4.01	80000.00

Insights

- Engineering: This department has an average performance score of approximately 4.01 and an average salary of $80,000.
- .HR: This department has a slightly higher average performance score of approximately 4.11 and an average salary of $81,818.18.
- Marketing: This department has the highest average performance score at approximately 4.17 but a lower average salary compared to HR, at $77,857.14.
- Sales: This department has the lowest average performance score at approximately 4.07 (note: the image shows 4.066, but the next digit is a 9, rounding up to 4.07) but the highest average salary at $82,068.96.

Actionable recommendations

Conduct a salary audit and implement a performance-based bonus structure to retain top talent.

Conclusions

KEY INSIGHTS

- There is a substantial human capital investment with total annual salary expenses of $4,850,000 for 60 employees
- 43.3% of employees earn above 80,000 while 56.7% earn below, requiring equitable compensation policies
- Marketing shows a negative correlation (-0.07) between performance and salary, contributing to high turnover (92.86)
- Engineering has the strongest (though generally weak) positive correlation (0.21), offering a model-for-pay Performance

ACTIONABLE RECOMMENDATIONS FOR HR

- Conduct comprehensive exit/stay surveys/interviews to evaluate leadership, resources and compensation Strengthen pay-for performance alignment and compensation equity across all department especially marketing
- Create clear career paths, while implementing targeted retention initiatives
- Foster a positive work environment and improve work life balance
- Build HR analytics platforms/dashboards to conduct regular monitoring and audits of compensation packages